# Effect on mortality of point-of-care, urine-based lipoarabinomannan testing to guide tuberculosis treatment initiation in HIV-positive hospital inpatients: a pragmatic, parallel-group, multicountry, open-label, randomised controlled trial

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## Summary

**Background** HIV-associated tuberculosis is difficult to diagnose and results in high mortality. Frequent extra-pulmonary presentation, inability to obtain sputum, and paucibacillary samples limits the usefulness of nucleic-acid amplification tests and smear microscopy. We therefore assessed a urine-based, lateral flow, point-of-care, lipoarabinomannan assay (LAM) and the effect of a LAM-guided anti-tuberculosis treatment initiation strategy on mortality.

**Methods** We did a pragmatic, randomised, parallel-group, multicentre trial in ten hospitals in Africa—four in South Africa, two in Tanzania, two in Zambia, and two in Zimbabwe. Eligible patients were HIV-positive adults aged at least 18 years with at least one of the following symptoms of tuberculosis (fever, cough, night sweats, or self-reported weightloss) and illness severity necessitating admission to hospital. Exclusion criteria included receipt of any anti-tuberculosis medicine in the 60 days before enrolment. We randomly assigned patients (1:1) to either LAM plus routine diagnostic tests for tuberculosis (smear microscopy, Xpert-MTB/RIF, and culture; LAM group) or routine diagnostic tests alone (no LAM group) using computer-generated allocation lists in blocks of ten. All patients were asked to provide a urine sample of at least 30 mL at enrolment, and trained research nurses did the LAM test in patients allocated to this group using the Alere Determine tuberculosis LAM Ag lateral flow strip test (Alere, USA) at the bedside on enrolment. On the basis of a positive test result, the nurses made a recommendation for initiating anti-tuberculosis treatment. The attending physician made an independent decision about whether to start treatment or not. Neither patients nor health-care workers were masked to group allocation and test results. The primary endpoint was 8-week all-cause mortality assessed in the modified intention-to-treat population (those who received their allocated intervention). This trial is registered with ClinicalTrials.gov, number NCT01770730.

**Findings** Between Jan 1, 2013, and Oct 2, 2014, we screened 8728 patients and randomly assigned 2659 to treatment (1336 to LAM, 1323 to no LAM). 108 patients did not receive their allocated treatment, mainly because they did not meet the inclusion criteria, and 23 were excluded from analysis, leaving 2528 in the final modified intention-to-treat analysis (1257 in the LAM group, 1271 in the no LAM group). Overall all-cause 8-week mortality occurred in 578 (23%) patients, 261 (21%) in LAM and 317 (25%) in no LAM, an absolute reduction of 4% (95% CI 1–7). The risk ratio adjusted for country was 0·83 (95% CI 0·73–0·96), p=0·012, with a relative risk reduction of 17% (95% CI 4–28). With the time-to-event analysis, there were 159 deaths per 100 person-years in LAM and 196 per 100 person-years in no LAM (hazard ratio adjusted for country 0·82 [95% CI 0·70–0·96], p=0·015). No adverse events were associated with LAM testing.

**Interpretation** Bedside LAM-guided initiation of anti-tuberculosis treatment in HIV-positive hospital inpatients with suspected tuberculosis was associated with reduced 8-week mortality. The implementation of LAM testing is likely to offer the greatest benefit in hospitals where diagnostic resources are most scarce and where patients present with severe illness, advanced immunosuppression, and an inability to self-expectorate sputum.

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## Introduction

Tuberculosis is the leading cause of death in people with HIV, accounting for about 360 000 deaths in 2013.[1,2] Case fatalities are significantly higher in individuals with tuberculosis plus HIV co-infection because of the increased disseminated extra-pulmonary tuberculosis, and more severe forms of disease associated with progressive immunosuppression.[1] Hospitals in endemic countries are overwhelmed with HIV-positive individuals with suspected tuberculosis who consume a large proportion of restricted

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## Research in context

### Evidence before this study

We searched PubMed for English-language studies published up to Sept 10, 2015, that examined the use of urine lipoarabinomannan testing (LAM) at the point of care and its effect on important patient outcomes. We combined search terms for urine LAM ("LAM", "lipoarabinomannan", "urine LAM", "urinary LAM", "LAM lateral flow assay", "LAM strip test", and "urine lipoarabinomannan") with those suggesting effect ("time to result", "time to diagnosis", "time to treatment", "morbidity", "mortality", and "outcome"). We identified eight published studies associating urine LAM positivity with poor prognostic features or increased mortality, or both. One study had assessed the use of urine LAM for monitoring treatment outcomes. Thus, previous studies have assessed the diagnostic accuracy of LAM and reported on the potential association between LAM positivity and prognostic features. However, in view of its moderate sensitivity and CD4 cell count-associated niche usefulness, whether LAM can reduce mortality in settings endemic for tuberculosis and HIV remained unclear.

### Added value of this study

Findings from our randomised trial suggest that a LAM-guided, prompt anti-tuberculosis treatment initiation strategy can reduce 8-week mortality in HIV-positive hospital inpatients.

### Implications of all the available evidence

Our data inform clinical practice and the roll-out of new diagnostic modalities in settings endemic for tuberculosis and HIV. The implementation of LAM testing is likely to offer the greatest benefit in hospitals where diagnostic resources are most scarce and where patients present with severe illness, advanced immunosuppression, and an inability to self-expectorate sputum. Collectively, the data suggest that policy makers should consider implementation of this low-cost bedside tuberculosis test in hospital inpatients in resource-limited settings with high tuberculosis and HIV burdens while further data accumulate.

health-care budgets. This consumption is due to several factors, including missed and delayed diagnoses of tuberculosis, which is a major challenge in individuals with HIV co-infection.[1] Indeed, post-mortem meta-analytical data suggest that tuberculosis is the cause of death in nearly 40% of HIV-positive patients, most being disseminated (>85%) and nearly half remaining undiagnosed at the time of death.[3] Severely ill patients with tuberculosis and HIV co-infection are often unable to self-expectorate sputum, or have extra-pulmonary disease thus needing sputum induction or alternative invasive sampling to acquire material for tuberculosis diagnostic testing.[4] Such resources are often unavailable in countries endemic for both tuberculosis and HIV.[5] Furthermore, patients with tuberculosis and HIV co-infection with advanced immunosuppression often have low bacillary loads in sputum and other body cavity fluids—eg, pleural and pericardial fluid, thus reducing the sensitivity of both traditional diagnostic tests (smear and culture) and newer nucleic acid amplification-based diagnostics (eg Xpert MTB/RIF [*Mycobacterium tuberculosis*/resistance to rifampacin]).[6–8] Because neither smear nor Xpert MTB/RIF provides immediate results at the point at which care is provided, a need remains for an affordable bedside diagnostic test for tuberculosis that can potentially minimise patient dropout, transmission, and mortality in hospital inpatients with advanced immunosuppression.

The Alere Determine tuberculosis lipoarabinomannan Ag lateral flow strip test (LAM; Alere, USA, www.alerehiv.com) is a commercially available bedside tuberculosis diagnostic instrument providing a result within 25 min, using about 60 µL of urine to detect lipoarabinomannan, a glycolipid antigen of the *M tuberculosis* cell wall.[9] The diagnostic accuracy of the LAM test has been moderate overall. The best performance has been noted among HIV-positive hospital inpatients with advanced immunosuppression (CD4 cell count ≤200 cells/µL).[10] In this subgroup, pooled sensitivity was 46% and specificity was 94%, respectively. Accurate assessment of the LAM-test specificity has been difficult in view of the limitations of diagnosis in patients with advanced immunosuppression and differences in LAM-test cutoffs used; with rigorous reference standard diagnosis and the updated cutoff, specificity is likely to be more than 95%.[11] The urine LAM test has shown incremental yield over sputum-smear microscopy[12,13] and identifies patients with the most severe illness and the highest bacillary load and mortality,[14,15] and the persistence of LAM positivity at 2 months after treatment initiation has been shown to be a poor prognostic indicator.[16]

The suboptimum accuracy of existing diagnostic methods in this subgroup of severely ill HIV-positive hospital inpatients means that empirical anti-tuberculosis treatment is common. Thus, whether, and to what extent, LAM testing might improve patient-important clinical outcomes or merely displace empirical treatment in true positive cases is unclear.[17] Indeed, large randomised controlled trials of Xpert MTB/RIF have not shown a significant effect on key clinical patient outcomes[18,19] because of the mitigating effect of empirical treatment practices.[20] We postulated that, despite moderate overall sensitivity, the addition of adjunctive LAM testing to available sputum-based tuberculosis diagnostics to guide anti-tuberculosis treatment initiation would improve tuberculosis treatment-related outcomes including mortality, morbidity, and length of hospital stay for HIV-positive hospital inpatients, compared with use of available sputum-based

tuberculosis diagnostics alone. Thus, we did a multi-country trial to assess these effects.

## Methods

### Study design and participants

We did a pragmatic, parallel-group, open-label, multicentre, randomised trial in ten urban or periurban hospitals in Africa—four in South Africa, two in Tanzania, two in Zambia, and two in Zimbabwe. The study was approved by the appropriate national regulatory authorities and by the ethics committee at each site. Additional methods are provided in the online appendix.

Patients admitted to the ten hospitals were screened for study inclusion. A detailed description of each hospital is in the appendix, including: (1) local figures for HIV and tuberculosis prevalence (where available); (2) the structure of routine clinical care; and (3) the availability of radiology services, sampling resources—eg, sputum induction facilities, and laboratory and tuberculosis diagnostics infrastructure (smear, Xpert MTB/RIF, and culture). Study research nurses and assistants did the screening in emergency units, short-stay wards, and medical wards at the hospitals on Monday to Friday during office hours (0800 to 1600), with the aim to enrol patients as soon as possible after admission. Thus, patients initiated on treatment outside these designated hours were excluded from enrolment. Provider-initiated counselling and HIV testing were offered as part of the screening process. The inclusion criteria included: (1) having HIV; (2) presenting with at least one of the following symptoms of tuberculosis: fever, cough, drenching night sweats, or self-reported weightloss;[21] (3) illness severe enough to necessitate admission to hospital; (4) aged at least 18 years; and (5) granting informed consent. Exclusion criteria included: (1) not having HIV; (2) receipt of any anti-tuberculosis medicine in the 60 days before testing; and (3) being unable to provide at least 30 mL of urine. All patients provided informed written consent in their first language.

### Randomisation and masking

We randomly assigned eligible patients (1:1) to receive either a urine LAM test plus standard available tuberculosis diagnostic tests (smear, Xpert MTB/RIF, and culture; LAM group) or standard tuberculosis diagnostic tests alone (no LAM group), using computer-generated allocation lists. Separate lists were generated for each country in blocks of ten. Once an eligible patient was identified, the research nurse at each site (who did not have access to these lists) either contacted a centrally located data manager by telephone to obtain assignment or used a text-message system, which automatically accessed the allocation list used to assign each patient. In the early part of the study, a few duplicate random number allocations occurred with the automated text system because of delayed responses with inconsistent cellphone and internet connectivity at specific sites. We quickly identified this difficulty and abandoned the text system. Neither patients or research nurses were masked to either allocation or test results.

### Procedures

At the time of enrolment, all patients (irrespective of group allocation) were asked by the research nurse to provide a spot urine specimen (≥30 mL) collected in a sterile container. The nurse obtained a spot catheter sample if a urinary catheter was in situ in bedbound patients. Patients were also asked to expectorate at least two sputum samples for routine tuberculosis diagnosis tests. For patients unable to self-expectorate, sputum-induction facilities were established at all hospital sites to improve sample acquisition and culture-based diagnosis.

See Online for appendix

For all patients allocated to the LAM group, the study research nursing staff did the urine LAM testing (using the Alere Determine TB LAM Ag lateral flow strip test) at the patients' bedside according to manufacturer's instructions, except in Zimbabwe where the test was done by state-certified laboratory technologists in

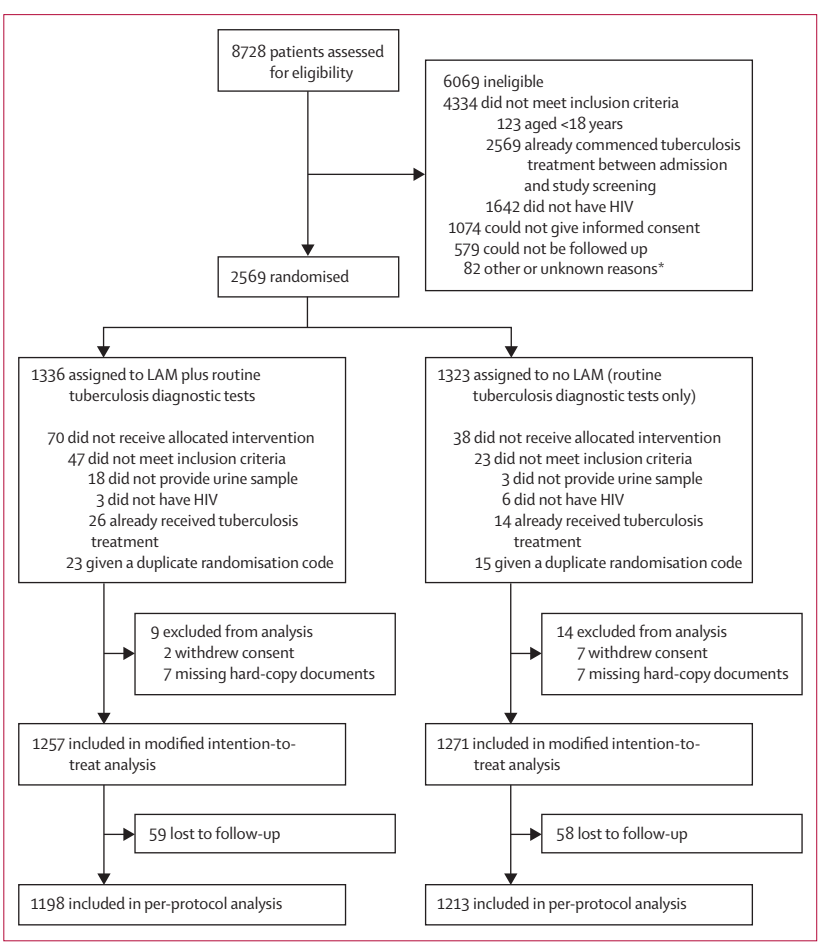



***Figure 1:*** **Trial profile**
*82 screened patients were excluded either for an unknown reason (eg, missing screening data) or because they were already enrolled in other studies.

accordance with the national regulations for new medical diagnostic tests. Further details of research nurse training, LAM-test secure storage, LAM performance, the reading of LAM tests, and the reference cards are in the appendix. The nurses who did the tests both documented the results in the patient folder, and verbally informed the attending clinicians either that "the LAM test is positive (grades 2–5)[11] and starting anti-tuberculosis treatment is recommended" or that "the LAM test is negative (grades 0–1); however, this does not exclude the diagnosis of tuberculosis". The grade 2 cutoff point deemed as positive in our study using the pre-January, 2014, reference card corresponds to the first positive band in the post-January, 2014, reference card (appendix). The attending physicians made independent decisions about whether to start treatment or not.

| | Overall (n=2528) | LAM (n=1257) | No LAM (n=1271) |
|---|---|---|---|
| Age (years) | 37 (30–44) | 37 (30–44) | 37 (30–44) |
| Sex | | | |
| Male | 1228 (49%) | 628 (50%) | 600 (47%) |
| Female | 1300 (51%) | 629 (50%) | 671 (53%) |
| Previous tuberculosis | 682 (27%) | 339 (27%) | 343 (27%) |
| Days between admission and study enrolment | 1 (1–2) | 1 (1–2) | 1 (1–2) |
| HIV status | | | |
| CD4 cell count (cells per μL) | 84 (26–208) | 81 (26–198) | 87 (28–217) |
| WHO clinical stage | | | |
| Stage 1 | 59 (2%) | 28 (2%) | 31 (2%) |
| Stage 2 | 671 (27%) | 337 (27%) | 334 (26%) |
| Stage 3 | 1331 (53%) | 675 (54%) | 656 (52%) |
| Stage 4 | 438 (17%) | 204 (16%) | 234 (18%) |
| ART at randomisation | 1224 (48%) | 604 (48%) | 620 (49%) |
| Co-trimoxazole prophylaxis at randomisation | 1130 (45%) | 552 (44%) | 578 (45%) |
| ART initiated by 8-week follow-up if alive and no ART at study entry | 615/848† (73%) | 312/435 (72%) | 303/413 (73%) |
| Tuberculosis symptoms | | | |
| Cough | 2350 (93%) | 1166 (93%) | 1184 (93%) |
| Weight loss | 2180 (86%) | 1084 (86%) | 1096 (86%) |
| Night sweats | 1894 (75%) | 927 (74%) | 967 (76%) |
| Fever | 1896 (75%) | 950 (76%) | 946 (74%) |
| Clinical severity markers | | | |
| Body-mass index (kg/m²)‡ | 18·8 (16·6–21·4) | 18·8 (16·7–21·4) | 18·7 (16·5–21·4) |
| Weight (kg)‡ | 51·0 (45·0–59·0) | 52·0 (45·5–59·9) | 51·0 (45·0–59·0) |
| Urea concentration (mmol/L)§ | 4·9 (3·2–8·8) | 4·8 (3·2–8·7) | 4·9 (3·2–9·3) |
| Respiratory rate (breaths per min) | 22 (20–26) | 22 (20–26) | 22 (20–26) |
| Heart rate (beats per min) | 100 (88–115) | 100 (87–114) | 100 (88–116) |
| Systolic blood pressure (mm Hg) | 110 (100–120) | 110 (100–120) | 110 (100–120) |
| Haemoglobin (g/L)§ | 92 (71–112) | 93 (72–113) | 90 (71–111) |
| Morbidity indices at baseline | | | |
| TBscore | 6 (5–8) | 6 (5–8) | 6 (5–8) |
| Karnofsky performance status | 50 (40–70) | 50 (40–70) | 50 (40–60) |

Data are median (IQR) or n (%). The modified intention-to-treat population were all participants who received their allocated intervention and were not excluded from analysis. LAM=lipoarabinomannan. ART=antiretroviral therapy. *Number of patients missing data for a particular variable: CD4 cell count: n=190. WHO clinical stage: n=29. ART initiation by 8 weeks: n=62. Respiratory rate: n=63. Heart rate: n=22. All other variables: <10 unless specified in footnotes below. †Denominator is patients alive and not on ART at enrolment. ‡Bed-bound patients were unable to be weighed in most cases. Consequently, 352 (14%) of 2528 weights are missing. Similarly, 295 (16%) of 2525 body-mass index data are missing because several heights were not measured. Thus, many patients (623/2528) also did not have tuberculosis score data because body-mass index data are necessary for calculation of this score. §524 and 692 patients are missing data for haemoglobin, and urea, respectively.

***Table 1:*** **Baseline characteristics of the modified intention-to-treat population***

A clinical assessment by the attending physicians and chest radiograph facilities were available in most cases (appendix). Additional radiological facilities—eg, ultrasound or CT scanning, and non-sputum sampling were differentially available in study hospitals, and requesting and reporting these investigations was at the discretion of the attending clinical team. Similarly, the attending clinical team made all decisions regarding patient therapy, and initiation and timing of anti-tuberculosis treatment, including acting upon the LAM-test results. If patients started anti-tuberculosis treatment, clinicians were asked for the primary reason motivating their treatment decision. WHO guidelines for the treatment of smear-negative tuberculosis were routinely used at study hospitals. If a patient was positive for a smear or Xpert MTB/RIF test, but the decision to initiate treatment was based on clinical findings and findings from a chest radiograph and made before the availability of this result, then the patient was deemed to have been treated based on clinical or radiographic evidence suggestive of tuberculosis.

Tuberculosis diagnostic tests were not standardised across study sites (appendix). Measurement of laboratory biochemistry was not mandatory as part of the study protocol. Consequently, data were only attained if requested by attending clinicians.

Study staff actively monitored hospital inpatients. Documentation about LAM results and sputum-based diagnostics were provided in the patients' folders. Details about post-discharge care and linkage are in the appendix. Research staff followed up patients at 8 weeks after randomisation (within a range of 14 days after this timepoint) for all patients discharged from hospital. Follow-up was done either in person, or if not possible, by telephone. Families, and hospital and local death registries, were contacted to retrieve information about mortality and dates of deaths. Attending clinicians were asked to categorise the cause of death as (1) likely to be tuberculosis, (2) possible tuberculosis, or (3) not tuberculosis and specify an alternative cause of death (appendix).

## Outcomes

The primary outcome was all-cause mortality at 8 weeks after randomisation, considering death both as a binary outcome at 8 weeks, and as a time-to-event outcome. Secondary outcomes were overall proportion of study patients initiated on anti-tuberculosis treatment and time-to-treatment initiation, changes in tuberculosis-related

morbidity (graded using the TBscore[22] and Karnofsky performance score[23] [see appendix]), diagnostic accuracy of the urine LAM test, and hospital length of stay from admission and enrolment.

### Statistical analysis

We calculated the sample size based on the following assumptions: tuberculosis prevalence and 8-week mortality in hospital inpatients with both HIV and tuberculosis with advanced immunosuppression are 50% and 25%,[24] respectively; rapid introduction of anti-tuberculosis drugs might reduce mortality by as much as 30%; and the loss to follow-up would be up to 10%. On the basis of these assumptions, we estimated that a sample size of 2650 patients would result in 80% power to detect a reduction in mortality of at least 25% with a two-sided type I error rate of 5%. The mortality-power calculation was based on the number of events needed for the entire cohort and the study was not powered to detect mortality differences between clinical and demographic subgroups. Our study was powered to assess both the primary and secondary outcomes (appendix).

We used descriptive statistics to characterise the study population by group. We used a modified intention-to-treat (modified ITT) and per-protocol approach (PPA) to assess the primary mortality outcome. When considering the

| | Sensitivity (n/N [%, 95% CI]) | Specificity (n/N [%, 95% CI]) | Positive likelihood ratio (95% CI) | Negative likelihood ratio (95% CI) |
|---|---|---|---|---|
| Overall | 156/342 (45·6% [40·4–50·9]) | 736/830 (88·7% [86·3–90·7]) | 4·03 (3·89–4·18) | 0·61 (0·61–0·62) |
| CD4 cell count subgroups† | | | | |
| >200 cells per μL | 7/49 (14·3% [7·1–26·7]) | 197/211 (93·4% [89·2–96·0]) | 2·15 (0·35–13·3) | 0·92 (0·88–0·96) |
| ≤200 cells per μL | 136/267 (50·9% [45·0–56·9]) | 495/569 (87·0% [84·0–89·5]) | 3·92 (3·76–4·08) | 0·56 (0·56–0·57) |
| >100 cells per μL | 20/103 (19·4% [12·9–28·0]) | 349/379 (92·1% [88·9–94·4]) | 2·45 (1·53–3·93) | 0·88 (0·85–0·90) |
| ≤100 cells per μL | 123/213 (57·7% [51·0–64·2]) | 343/401 (85·5% [81·8–88·6]) | 3·99 (3·82–4·18) | 0·49 (0·48–0·51) |
| >50 cells per μL | 50/170 (29·4% [23·1–36·7]) | 459/500 (91·8% [89·1–93·9]) | 3·59 (3·11–4·13) | 0·77 (0·76–0·78) |
| ≤50 cells per μL | 93/146 (63·7% [55·6–71·1]) | 233/280 (83·2% [78·4–87·1]) | 3·80 (3·60–4·00) | 0·44 (0·42–0·45) |

LAM=lipoarabinomannan. MTB=*Mycobacterium tuberculosis*. RIF=resistance to rifampicin. *The reference standard used was any sputum or non-sputum sample that was positive for tuberculosis culture or Xpert MTB/RIF tests. For most patients, reference testing was restricted to one sputum culture or Xpert MTB/RIF test, which, in the context of patients with HIV co-infection and advanced immunosuppression, might underdiagnose tuberculosis, introducing sampling bias, and potentially lead to an underestimation of LAM specificity. The role of non-tuberculous mycobacteria affecting specificity is unclear. Only patients with urine LAM testing were eligible (n=1257); the following patients were excluded from this analysis: 75 from LAM group not able to provide any samples for reference diagnosis; seven not having at least one Xpert MTB/RIF or liquid tuberculosis culture on either a sputum or non-sputum sample; and three having an invalid LAM despite repeat testing. 1172 were included in the diagnostic accuracy analysis. Only urine LAM grade 2 or more was considered test positive (pre January, 2014, reference card). †95 patients were missing CD4 cell counts in the LAM group and were thus excluded from subgroup diagnostic accuracy analyses.

***Table 2:*** **Diagnostic accuracy of urine LAM test***

| | LAM | No LAM | Absolute risk reduction (% [95% CI]) | Relative risk reduction (% [95% CI]) | Adjusted risk ratio* (95% CI), p value | Adjusted hazard ratio† (95% CI), p value |
|---|---|---|---|---|---|---|
| All cause 8-week mortality (MITT) | 261/1257 (20·8%) | 317/1271 (24·9%) | 4% (1–7) | 17% (4–28) | 0·83 (0·73–0·96), 0·012 | .. |
| Number of deaths | 279 | 334 | .. | .. | .. | .. |
| Person-years at risk | 175 | 170 | .. | .. | .. | .. |
| Mortality per 100 person-years at risk | 159 (141–179) | 196 (176–218) | .. | .. | .. | 0·82 (0·70–0·96), 0·0145 |
| All cause 8-week mortality (PPA) | 261/1198 (21·8%) | 317/1213 (26·1%) | 5% (1–8) | 17% (4–28) | 0·83 (0·72–0·96), 0·011 | .. |
| Number of deaths | 279 | 334 | .. | .. | .. | .. |
| Person-years at risk | 175 | 170 | .. | .. | .. | .. |
| Mortality per 100 person-years at risk | 160 (142–180) | 197 (177–219) | .. | .. | .. | 0·82 (0·70–0·96), 0·0134 |
| Excluding non-tuberculosis deaths‡ | 225/1216 (18·5%) | 270/1221 (22·1%) | 4% (0–7) | 16% (2–29) | 0·84 (0·71–0·98), 0·025 | .. |
| Number of deaths | 239 | 284 | .. | .. | .. | .. |
| Person-years at risk | 173 | 168 | .. | .. | .. | .. |
| Mortality per 100 person-years at risk | 138 (122–157) | 169 (151–190) | .. | .. | .. | 0·82 (0·69–0·98), 0·0272 |

Data are n/N (%) or n (95% CI) unless otherwise stated. MITT=modified intention-to-treat analysis with all lost to follow-up included and censored at last date alive in the study, which was the hospital-discharge date (n=2528). PPA=per-protocol analysis with lost to follow-up excluded from the analysis (117 patients lost to follow-up: 58 no LAM, 59 LAM, thus n=2411). MTB=*Mycobacterium tuberculosis*. RIF=resistance to rifampicin. *Risk ratios presented were adjusted by country of enrolment because of stratified randomisation. †Hazard ratios presented were adjusted by country of enrolment because of stratified randomisation. The overall interaction between study group and country of enrolment was not significant (p=0·370) and thus not included in the model for adjustment of hazards. ‡Attending clinicians or death certification classified deaths as tuberculosis, possible tuberculosis, or non-tuberculosis. All deaths of patients with a positive sample from an Xpert MTB/RIF test or tuberculosis culture test were deemed as tuberculosis-related deaths. This analysis excluded deaths classified as non-tuberculosis and having negative results from Xpert MTB/RIF or tuberculosis culture tests (n=83/578).

***Table 3:*** **Effects of urine LAM-guided initiation of tuberculosis treatment on 8-week all-cause mortality**

primary outcome as a binary outcome at 8 weeks, we analysed it at 56 days, calculating proportions in each study group and risk ratios (RR) with 95% CIs using the mid-P exact method. Considered as a time-to-event outcome, we analysed the primary outcome as a rate using time-to-event analysis censored at 70 days (ie, 56+14 days) to identify the event rate per person-year and hazard ratios (HRs) with 95% CIs. We did a sensitivity analysis to compare PPA and modified ITT approaches considering death as a binary outcome at 8 weeks. We constructed time-to-event curves by means of product-limit estimation and used a log-rank test (stratified by country to account for randomisation by country) to compare the survival distributions of the study groups. We measured the intervention effect both as the relative risk of death in the LAM group versus the no LAM group using binary regression adjusted for country of enrolment, and the HR of LAM versus no LAM using a Cox proportional hazards model with group and country as the main effects and reporting 95% CIs for all analyses. We tested proportional hazards assumptions using Schoenfeld residuals and in all presented models proportional hazards assumptions were met (p>0·05). We used the $\chi^2$ test to compare the proportion of patients starting anti-tuberculosis treatment and the diagnostic-test combinations used. We used a Mann-Whitney test to compare differences in TBscore and days to initiation of anti-tuberculosis treatment. We also examined the effects of LAM testing on the primary and secondary outcomes in predefined subgroups (CD4 cell count terciles [0–50, 51–100, and >100 cells per μL], CD4 cell count binary strata [>200, ≤200, >100, ≤100, >50, ≤50 cells per μL], HIV clinical subgroups [WHO clinical stages 1 and 2, WHO clinical stages 3 and 4], and countries [South Africa, Zimbabwe, Zambia, and Tanzania]). Details regarding subgroup analysis, interaction between variables, and adjustment for interim analysis are in the appendix. We did the statistical tests with Stata version 11. A single planned interim efficacy review was done by a data and safety monitoring board when 13% of the intended participants had completed their 8-week follow-up, to review the trial for appropriateness to continue. For this early look, 0·00016 of α was spent with little effect on the reported p values. This trial is registered with ClinicalTrials.gov, number NCT01770730.

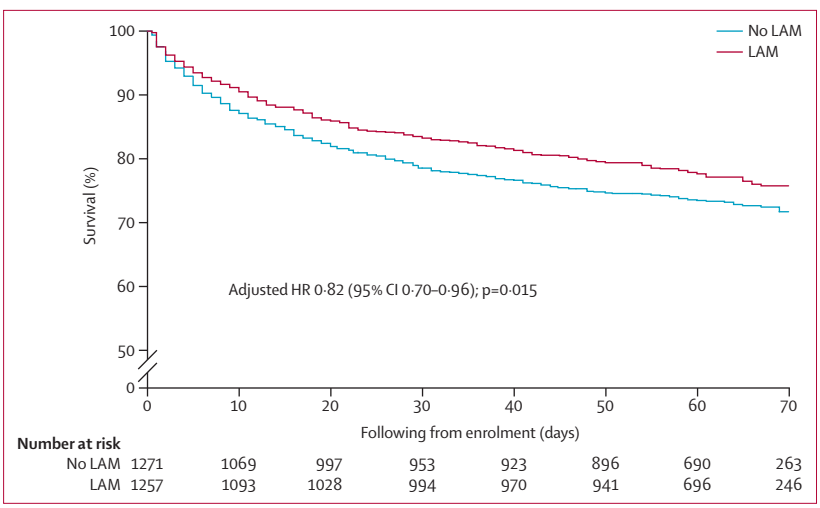



*Figure 2:* **Time to 8-week all-cause mortality**
HR=hazard ratio. LAM=lipoarabinomannan. Data are overall HRs and p values for study groups adjusted for country.

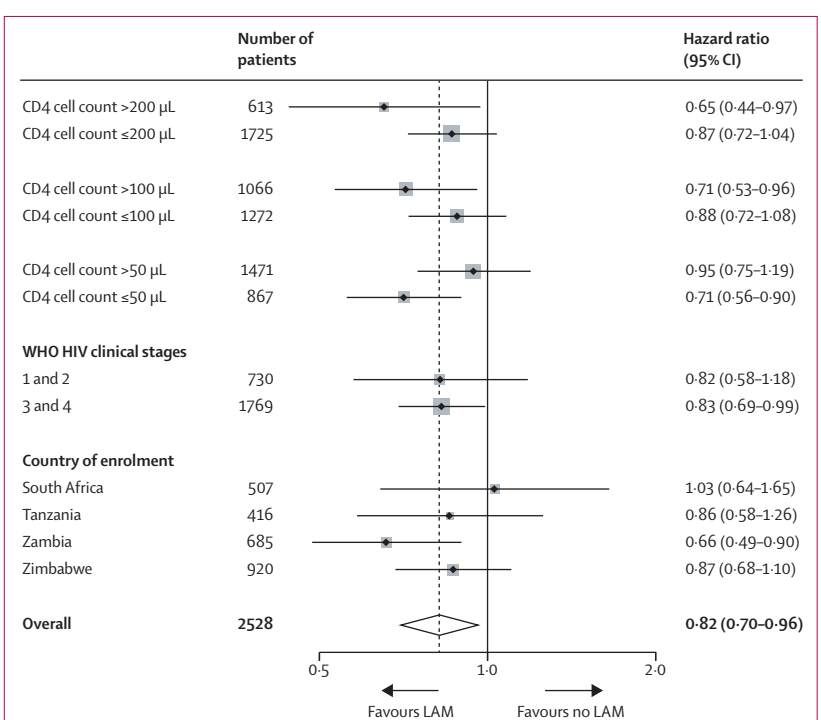



*Figure 3:* **Effect sizes for 8-week all-cause mortality**
LAM=lipoarabinomannan. Data are shown for clinical and country-specific subgroups. For each subgroup, the point estimate of the hazard ratio and 95% CI is shown. Number of deaths are shown in the appendix. Lines or CI not including 1 show p values <0·05 when viewed against the relevant baseline.

## Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The first author and the corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

## Results

Between Jan 1, 2013, and Oct 2, 2014, we screened 8728 possible participants (figure 1). Most were screened 4–48 h after hospital admission, yet by this time 2569 (29%) had already been empirically started on anti-tuberculosis treatment and were excluded. An additional 1074 patients (12%) were too ill to provide informed consent and were thus also excluded from enrolment. After screening, we assigned 2659 eligible patients to study interventions, and included 2528 in the modified ITT analysis (1257 in the LAM group, 1271 in the no LAM group). 8-week outcome data were available for 2411 (95%) of 2528 patients. Median

| | LAM (n=1257) | Deaths in LAM | No LAM (n=1271) | Deaths in no LAM | p value* |
|---|---|---|---|---|---|
| Patients starting anti-tuberculosis treatment | 648 (52%) | 131/648 (20%) | 598 (47%) | 141/598 (24%) | 0·024 |
| Patients not initiated on anti-tuberculosis treatment | 609 (48%) | 130/609 (21%) | 673 (53%) | 176/673 (26%) | 0·024 |
| Time to anti-tuberculosis treatment initiation (days) | | | | | |
| All countries | 0 (0–2) | | 1 (0–3) | | <0·0001 |
| South Africa | 1 (0–2) | | 1 (0–3) | | 0·043 |
| Tanzania | 0 (0–2) | | 1 (0–2) | | 0·009 |
| Zambia | 0 (0–2) | | 1 (0–3) | | 0·009 |
| Zimbabwe | 0 (0–2) | | 0 (0–4) | | 0·017 |
| Cumulative proportion receiving anti-tuberculosis treatment for patients who started therapy by the end of: | | | | | |
| Day 1† | 357/648 (55%) | 82/357 (23%) | 241/598 (40%) | 58/241 (24%) | <0·0001 |
| Day 2 | 461/648 (71%) | 100/461 (22%) | 362/598 (61%) | 84/362 (23%) | 0·0001 |
| Day 3 | 513/648 (79%) | 110/513 (21%) | 413/598 (69%) | 95/413 (23%) | <0·0001 |
| Day 4 | 544/648 (84%) | 113/544 (21%) | 452/598 (76%) | 109/452 (24%) | 0·0002 |
| Day 8 | 590/648 (91%) | 121/590 (21%) | 535/598 (89%) | 129/535 (24%) | 0·35 |

Data are n (%) or median (IQR). LAM=lipoarabinomannan. *p values are for the comparison of times to treatment between study groups. †Day of admission. Days 1, 2, 3, 4, and 8 were days during which results of tuberculosis investigations would normally be returned.

***Table 4:*** **Anti-tuberculosis treatment**

age and gender did not differ between screened but excluded and enrolled patients (data not shown). The baseline admission characteristics of patients were balanced between the study groups (table 1). Differences in clinical status were noted between countries (appendix).

Use of tuberculosis diagnostic tests and treatment were dictated by the availability and success of sampling, local capacity, and attending clinician practice; nonetheless, nearly all patients received a chest radiograph, and at least one tuberculosis culture or Xpert MTB/RIF test (appendix), and only 173 (7%) were unable to provide a sample for tuberculosis diagnostic testing (appendix), balanced between study groups. By contrast, only 21 (<1%) patients were unable to provide a urine sample within 24 h of inclusion (figure 1). Some differences in use of routine diagnostic tests were noted between countries (appendix). No differences were noted between groups for chest radiographs suggestive of tuberculosis, and smear or Xpert MTB/RIF-positive sputum or non-sputum samples (appendix). We did urine LAM testing on all patients in the LAM group, and only three had a test classified as invalid after a repeat LAM strip failed. 274 (22%) of 1254 LAM tests were positive (grades 2–5). LAM had an overall sensitivity of around 46% and specificity of around 90% (table 2). The highest sensitivity was in patients with CD4 cell counts of 50/μL or fewer (table 2).

Overall all-cause 8-week mortality occurred in 578 (23%) of 2528 patients, 261 (21%) of 1257 in the LAM group and 317 (25%) of 1271 in the no LAM group, an absolute reduction of 4% (95% CI 1–7; table 3). 301 (52%) of 578 deaths occurred during the study hospital admissions. The risk ratio adjusted for country was 0·83 (95% CI 0·73–0·96), p=0·012, with a relative risk reduction of 17% (95% CI 4–28). With the time-to-event analysis, there were 159 deaths per 100 person-years in LAM and 196 per 100 person-years in no LAM (hazard ratio for LAM adjusted for country 0·82 [95% CI 0·70–0·96], p=0·0145; figure 2, table 3). The mortality effect was heterogeneous across sites (figure 3; appendix). The sensitivity analysis comparing the primary outcome in the per-protocol and modified ITT populations did not change overall study outcomes (appendix).

The effect size of LAM testing on mortality by subgroup varied by country and clinical characteristics (figure 3, appendix), but the greatest number of deaths and mortality reduction occurred in the subgroup with a CD4 cell count of 50/μL or fewer (264 [46%] of 578 deaths; appendix). We also noted differences in the effect of LAM testing between countries (figure 3, appendix). Inclusion of other possible confounders of mortality in addition to country, such as age at admission, CD4 cell count, the ability to obtain sputum-based diagnostic samples, time on antiretroviral therapy or co-trimoxazole prophylaxis before enrolment, Karnofsky performance status at enrolment, and the performance of Xpert MTB/RIF in a multivariate analysis, slightly improved the overall effect size of LAM testing on mortality (appendix).

The overall percentage of patients started on anti-tuberculosis treatment was higher in the LAM group than the no LAM group (p=0·024; table 4). The time-to-treatment initiation in the LAM group was shorter than in the no LAM group, with a higher proportion of those treated having anti-tuberculosis treatment initiated on days 0–3 (the first 96 h; table 4). Attending clinicians acted on LAM test results as the main reason for initiating anti-tuberculosis treatment (appendix). Adjunctive LAM improved time-to-treatment initiation in all countries. The use of adjunctive LAM to guide anti-tuberculosis treatment decision making led to fewer patients starting anti-tuberculosis treatment based on clinical or chest

| | LAM | No LAM | p value |
|---|---|---|---|
| Hospital length of stay from admission* (days) | 5 (3–9) | 5 (4–9) | 0·283 |
| Hospital length of stay from enrolment* (days) | 4 (2–7) | 4 (2–7) | 0·244 |
| Improvement in TBscore in patients receiving tuberculosis treatment† | 5 (3–6) | 5 (3–6) | 0·637 |
| Improvement in TBscore in patients with a positive tuberculosis culture test receiving treatment‡ | 5 (3–6) | 5 (3–6) | 0·311 |
| Change in Karnofsky performance score in patients receiving tuberculosis treatment§ | 30 (10–40) | 30 (10–40) | 0·811 |
| Change in Karnofsky performance score in patients with a positive tuberculosis culture test receiving treatment§ | 30 (10–40) | 20 (10–40) | 0·306 |

Data are median (IQR) improvements in scores between baseline and 8-week follow-up. *N=2222/2528. 302 inpatient deaths and four deaths missing dates of hospital discharges were excluded from this analysis. †N=548/904 (291 in LAM and 257 patients in no LAM) patients alive and initiated on anti-tuberculosis treatment included in analysis. 356 (191 in LAM and 165 in no LAM) missing complete tuberculosis score at baseline, follow-up, or both, because of (1) inability to calculate baseline body-mass index because weighing bed-bound patients was not possible in most cases, or (2) 8-week follow-up was only possible by telephone and thus neither body-mass index or mid-upper arm circumference were measurable at follow-up. ‡267 (136 LAM patients in LAM and 131 in no LAM) of 427 patients with a positive tuberculosis culture were alive and initiated on anti-tuberculosis treatment. 160 (87 LAM patients in LAM and 73 in no LAM) missing a complete tuberculosis score at baseline, follow-up, or both. ‡412 (217 in LAM and 195 patients in no LAM) of 427 patients with a positive tuberculosis culture alive and initiated on anti-tuberculosis treatment. 15 (six in LAM and nine patients in no LAM) missing complete Karnofsky perfomance scores at baseline, follow-up, or both. §867 (465 in LAM and 402 in no LAM) of 904 patients alive and initiated on anti-tuberculosis treatment included in the analysis. 37 (17 in LAM and 20 in no LAM) missing complete Karnofsky performance scores at baseline, follow-up, or both.

***Table 5:*** **Effects of urine LAM testing on secondary outcomes**

radiographic evidence alone or a positive smear/Xpert MTB/RIF sample, although this effect differed between countries (appendix). The main reasons why clinicians did not initiate anti-tuberculosis treatment at enrolment for those eventually treated for tuberculosis were (1) another primary diagnosis was favoured (eg, bacterial pneumonia), or (2) they were awaiting the results of diagnostic investigations (appendix).

Table 4 (and appendix) shows that mortality was lower in the LAM group than the no LAM group overall in both patients who were given anti-tuberculosis treatment and non-treated patients, and for each additional day of treatment initiation after enrolment. However, the only significant reduction occurred in the non-treatment group (appendix).

No differences were noted between study groups for other secondary outcomes (table 5, appendix). The case fatality ratio was 0·12 deaths per admissions (301/2528). No adverse events were associated with LAM testing.

## Discussion

Adjunctive urine LAM tests reduced all cause 8-week mortality in hospital inpatients with HIV, symptoms of tuberculosis, and advanced immunosuppression. Use of LAM tests ensured that more patients initiated anti-tuberculosis treatment and this occurred earlier than patients not receiving LAM testing.

This is the first trial of any tuberculosis diagnostic test to show a mortality benefit. By contrast, two large diagnostic trials[18,19] of the Xpert MTB/RIF assay did not show either morbidity or mortality benefit, despite a reduction in the time-to-treatment initiation. The effects of LAM-guided anti-tuberculosis treatment are likely because of more patients starting tuberculosis therapy, and earlier initiation of anti-tuberculosis treatment. However, the dominant driver seems to be a large overall proportion of patients being initiated on treatment. This finding is consistent with our previous findings that LAM identifies patients with possible tuberculosis not detected by clinicians who make routine empirical treatment decisions.[17] Thus, despite only having 50% to 70% sensitivity, depending on CD4 cell count, bedside LAM testing improves the sensitivity of clinical and chest radiograph diagnosis, and leads to rapid, life-saving, treatment initiation in the most susceptible patient subgroups, and at low cost.[13] Urine LAM testing has no infrastructural requirements and a cost of only US$2·66 per test (as of January, 2015).

Several population-specific and test-specific factors help with earlier and expanded diagnosis, and treatment based on adjunctive urine LAM testing, which are likely to have decreased mortality. First, the test is not dependent on the availability of sputum or the presence of bacteria in the sample; both troublesome factors when diagnosing patients with low CD4 cell counts with conventional rapid tests (smear microscopy or Xpert MTB/RIF). Thus, LAM testing might result in the treatment of tuberculosis cases missed by empirical treatment or available diagnostics. Second, event rates in people with advanced immunosuppression (CD4 cell count ≤50/μL) in different settings are very high[25-27] and disseminated tuberculosis in such patients might even result in a clinical syndrome resembling septic shock.[28] Thus, reducing time-to-treatment might also offer benefit in this patient group. Indeed, more patients received anti-tuberculosis treatment in the first 4 days in the LAM group, and LAM reduced mortality by 29% (calculated from HRs) in patients with a CD4 cell count of 50/μL or fewer. The same CD4 cell count cutoff point, underpinned by high event rates, has previously been reported as significant for reduced mortality in trials investigating the rapid uptake of antiretroviral therapy.[27]

The mortality effect was heterogeneous across sites (reduced at three sites) but only reached statistical significance at the Zambian site. However, although randomisation was stratified by site, our study was not

powered to detect site-specific mortality differences, and therefore the absence of significant differences at all sites was not surprising. Finding such differences is not unexpected and shows the heterogeneity of clinical practice and health-care systems. In Zambia, for example, patients were more likely to have poor prognostic features (WHO stage 4 HIV-associated disease, lower bodyweights, haemoglobin, and Karnofsky performance statuses, and higher TBscores) and sputum-based sampling success was lower, with less use of rapid sputum-based diagnostics to guide treatment decisions. By contrast, in South Africa, overall mortality was much lower, a greater proportion of patients were on antiretroviral drugs at enrolment, sampling was better, and a longer time occurred between hospital admission and study enrolment. All these factors could have diminished the effect of LAM testing in South Africa. Moreover, in different health-care settings, delays in patients with severe illness presenting to hospitals could be highly variable, attending clinicians might have very different probabilities of investigating patients, and the initiation of empirical treatment might depend on local practices.

The overall findings have important implications because many high tuberculosis-burden settings exist, with poor access to sputum induction facilities, a large undiagnosed HIV burden.[29] Furthermore, in a study[4] of hospital inpatients, only just more than half were able to provide sputum at any specific point during admission to hospital, lending support to the need for urine-based diagnostic approaches. In view of the estimated 300 000 HIV-positive deaths from tuberculosis in Africa, implementation of bedside urine LAM testing for HIV-positive hospital inpatients with suspected tuberculosis might save thousands of lives annually.[2] Thus, although the effect size is slight overall, the overall attributable effect on mortality is likely to be substantial in view of the magnitude of patients with advanced immunosuppression. National tuberculosis programmes in African settings with a high burden of both tuberculosis and HIV should urgently consider incorporation of hospital-based LAM testing for HIV-positive inpatients with possible tuberculosis co-infection. Additionally, WHO has assessed urine LAM testing, and this study supports the endorsement of LAM testing for routine tuberculosis diagnosis among hospital inpatients with HIV in high-incidence settings.

By contrast with mortality, secondary outcomes including hospital length of stay and tuberculosis-related morbidity were not improved by adjunctive LAM. The likely reason is survival bias, because among the surviving patients with immunosuppression and several comorbidities, clinical recovery was not significantly affected by a short difference in time-to-treatment. Likewise, hospital length of stay might have other more important drivers such as the pressure to free up beds, and local policies. Other important potential benefits of a LAM-based strategy that need further study include its usefulness for reducing invasive diagnostic sampling and radiological testing (hence saving costs) and a reduction in the inappropriate use of antibiotics that are often empirically given.

Our study had several limitations. A few patients received duplicate randomisation numbers because of a problematic text-based allocation system that was soon abandoned. However, this difficulty was identified early to eliminate any possibility of bias, by ensuring that patients were correctly allocated to their first text number and group, and duplicate numbers removed. Tuberculosis diagnostic tests and treatment, with the exception of sputum induction facilities, were based on local availability and at the discretion of attending clinicians. Sampling, the use of Xpert MTB/RIF, smear microscopy, and culture methods also varied across study sites because of local policy. Furthermore, illness severity varied across sites and likely underpinned many of the differences between site-specific outcomes. All these factors might have biased our findings. However, although generating difficulties with pooled effects, no significant interaction between study group and country was shown. We believe that this heterogeneity (between diagnostic tests and methods, treatment, and illness severity) emphasises the pragmatic nature of our study allowing contextual differences in effect to emerge, thus improving external validity and making us better able to address the question of where LAM testing might be most useful in a real-world setting. Additionally, our study more accurately shows the reality of routine clinical practice and the heterogeneous tuberculosis-diagnostic capacities between hospitals and countries. This was not a diagnostic accuracy study, thus reference standard testing was suboptimum and probably underestimated LAM-test specificity that in other studies with a robust reference has been shown to be highly specific (>95%).[30] Stratified CD4 cell count-specific randomisation from the outset was not undertaken because this was not expected to be different between sites and most patients were expected to have a CD4 cell count of fewer than 100/μL. Moreover, additional multivariate Cox proportional hazards modelling was done to explore for other potential confounders, thereby further strengthening the robustness of our conclusions.

Many patients did not meet the entry criteria for our study. In routine clinical practice, LAM testing would be available 24 h per day; however, logistical reasons, including scarce research budgets, meant that clinical study teams were not employed during the night and weekends to do LAM testing. Consequently, several individuals who might have been eligible for inclusion might have already initiated anti-tuberculosis treatment before screening and were thus excluded. Additionally, some patients were only screened 24–48 h after admission. A sensitivity analysis revealed similar basic demographics between the excluded and enrolled

patients, thus decreasing the likelihood of significant selection bias. Many severely ill patients were also incapable of providing informed consent, but whether LAM testing could have been of benefit in this patient group is uncertain. Exclusion of these patients might have led to either an overestimation or underestimation of the LAM effect on mortality—we speculate that their exclusion resulted in an underestimation of LAM effect because LAM testing has higher sensitivity among severely ill patients with high bacillary loads.[31]

Not all patients with a positive LAM result were immediately initiated on anti-tuberculosis treatment and this delay might also have reduced any potential effect on mortality, leading to further underestimations of the effects of LAM testing. However, in most cases the LAM results were acted on in a timely manner, suggesting that in these instances clinicians felt confident to initiate anti-tuberculosis treatment based on LAM test results. The only significant reduction in mortality occurred in the non-treatment group, which suggests the main benefit of LAM-guided therapy was the initiation of a greater overall proportion of patients, although earlier treatment might also play a part. The absence of a sham-like intervention or result or standardised reporting of sputum-based tuberculosis assays might have affected results either in favour of LAM, because of a clear recommendation for treatment initiation, or against LAM because of the documentation of a negative tuberculosis test result in the notes. In a diagnostic study such as this one, a sham result in control patients was impossible because it would have affected treatment decisions. However, any effect was probably negligible because the study staff actively monitored all the test results for both study groups, with folder notification irrespective of whether the result pertained to LAM, smear microscopy, or Xpert MTB/RIF testing.

In conclusion, urine LAM testing reduced 8-week mortality when used as a rapid adjunct to available tuberculosis diagnostics in hospital inpatients with HIV. LAM testing reduced mortality by enabling a greater proportion of patients to start treatment, and faster initiation of anti-tuberculosis treatment among the most ill and susceptible patients. The implementation of LAM testing is likely to offer the greatest benefit in hospitals where diagnostic resources are most limited, and where patients often present with severe illness, advanced immunosuppression, and an inability to self-expectorate sputum. Our study supports the use of LAM testing in all patients admitted to hospital with any one of four tuberculosis-related symptoms (fever, cough of any duration, weight loss, and night sweats) and that the test should be done as close to admission as possible. Although this is the first controlled study to assess mortality, the findings presented here suggest that national tuberculosis programmes in resource-limited settings with high HIV-associated tuberculosis burden should consider implementation of this low cost, rapid, bedside (or point-of-care) tuberculosis test while further operational data accumulate to confirm these findings.

#### Contributors

JGP, LSZ, DC, PC, AR, MH, PM, GT, and KD were involved in the conception and design of the study. KD obtained the funding. JGP, LSZ, DC, PC, PG, NM, GC, GK, AC, NL, CM, BM, HM, AR, MH, PM, and KD were involved in study implementation. JGP, ML, CJL, and KD did the analysis. ML, CJL, and TB were the study statisticians. JGP, LSZ, DC, PC, ML, CJL, MH, AR, GC, GT, and KD interpreted the data. JGP, LSZ, DC, PC, and KD wrote the first draft. All authors read and approved the manuscript.

#### Declaration of interests

KD has obtained speaker fees at industry-sponsored symposia and non-financial support from Alere in the form of kits and test strips, outside the submitted work. No other authors declare competing interests.

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